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Speciesist language and nonhuman animal bias in English Masked Language Models

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ABSTRACT

Warning: This paper contains examples of offensive language, including insulting or objectifying expressions.

Various existing studies have analyzed what social biases are inherited by NLP models. These biases may directly or indirectly harm people, therefore previous studies have focused only on human attributes. However, until recently no research on social biases in NLP regarding nonhumans existed. In this paper,[1] we analyze biases to nonhuman animals, i.e. speciesist bias, inherent in English Masked Language Models such as BERT. We analyzed speciesist bias against 46 animal names using template-based and corpus-extracted sentences containing speciesist (or non-speciesist) language. We found that pre-trained masked language models tend to associate harmful words with nonhuman animals and have a bias toward using speciesist language for some nonhuman animal names. Our code for reproducing the experiments will be made available on GitHub.[2]

1. Introduction

Recently, in the field of Natural Language Processing (NLP), Masked Language Models (MLMs) using Transformers (Vaswani et al., 2017), such as BERT (Devlin, Chang, Lee, & Toutanova, 2019) and RoBERTa (Liu et al., 2019), widely contributed to the state-of-the-art methods in downstream tasks. However, existing studies suggest that these models inherit social biases (Blodgett, Barocas, Daumé III, & Wallach, 2020; Sun et al., 2019). Here, social bias is defined as the systematic, unequal treatment based on one's social attribute (cf. Stanczak & Augenstein, 2021; Sun et al., 2019). Such biases cause differences in accuracy between majority and minority attributes (e.g. Romanov et al., 2019) and negative generalizations, e.g. in text generation (Garimella et al., 2021; Liu et al., 2020; Sheng, Chang, Natarajan, & Peng, 2019, 2021).

The studies of social bias in NLP target gender (e.g. Bolukbasi, Chang, Zou, Saligrama, & Kalai, 2016; Caliskan, Bryson, & Narayanan, 2017), race (e.g. Manzini, Yao Chong, Black, & Tsvetkov, 2019), religion and ethnicity (e.g. Li, Khashabi, Khot, Sabharwal, & Srikumar, 2020) and so on, all of which assume human attributes. However, to the best of the authors' knowledge, there are no similar bias studies on **nonhuman animals**.

In this paper, we use templates, corpus-extracted sentences and pre-trained MLMs to investigate if the bias regarding nonhuman, i.e. **speciesist bias**, is inherent in MLMs trained on English corpora.

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[1] Anonymous previous version of this paper is accessible at https://openreview.net/forum?id=dfqMpjZOgv4, we have also published a paper on this topic in Japanese (Takeshita, Rafal, & Araki, 2021)

[2] https://github.com/Language-Media-Lab/speciesist-language

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